

Chapter 32: The Newborn — Revision Notes

Williams Obstetrics 26e, pp. 1512–1544 of the PDF. High-yield numbers are in **bold**. The five tables (32-1 to 32-5) are transcribed from the PDF images (they are missing from the extracted chapter text). All 5 figures are included (the resuscitation algorithm, Fig 32-2, is printed twice, on p1518 and p1519; the p1511 bag-and-mask drawing is the Section 9 title-page art, not a numbered figure). The acid-base equations, lost from the extracted text, are restored from the PDF pages.

Big picture: Every birth needs **at least one person whose only job is the newborn** and who can give **positive-pressure ventilation (PPV)**, the single most important step in resuscitation. After birth: Apgar and cord gases describe the newborn's condition; then come the preventive care bundle (eyes, hepatitis B, vitamin K, screening) and routine care.

1. Attendants at Delivery

- **AAP/ACOG:** ≥1 qualified person at every birth, skilled in initial steps and PPV, **responsible only for the newborn** (pediatrician, NP, anesthesia provider, or trained nurse). If absent, it falls to the **obstetrical attendant**.
- **High-risk deliveries:** a full-resuscitation team **present**, and **immediately available** for every resuscitation (**not on call from home** or a remote area).
- Team **simulation training** recommended.

2. Transition to Air Breathing

- Placental → pulmonary gas exchange: **PVR falls**, pulmonary flow rises, and the **ductus arteriosus and foramen ovale** begin to close.
- **Lung aeration itself triggers the cardiovascular changes** at birth.

Lung fluid clearance mechanisms

1. **Fetal adrenaline surge** late in labor → epithelium switches from secreting to **reabsorbing** fluid (sodium-channel activation). Probably **minor** (blockade delays but doesn't prevent clearance).
2. **Mechanical forces:** thoracic compression can expel up to $\frac{1}{3}$ of lung liquid. **Uterine contractions changing fetal posture** (early in labor) matter more than the "vaginal squeeze".
3. **After birth (major):** most aeration happens during **inspiration within 3–5 breaths**; the transpulmonary gradient pushes fluid into the interstitium, then it is cleared by blood and lymphatics. **No clearance between breaths.** Without PEEP, fluid can re-enter airspaces in expiration → may contribute to **transient tachypnea of the newborn**.
4. **Surfactant** (type II pneumocytes) ↓ surface tension and prevents alveolar collapse. Deficiency (preterm) → **RDS**.

Breathing mechanics

- First breaths need **high negative intrathoracic pressure**; opening pressure falls with each breath. By about the **5th breath** pressure-volume changes resemble the adult's.
- Air replaces fluid → ↓ vascular compression → ↓ PVR → ↓ pulmonary artery pressure → **ductus closes**.

Why delayed cord clamping helps

- In utero, **umbilical venous return is the main LV preload** (pulmonary flow is tiny).
- **Clamping before lung aeration** → ↓ preload → ↓ cardiac output → **bradycardia**.
- **Clamping after aeration** → smoother transition, no fall in output → interest in **physiological cord clamping** (special resuscitation trolleys; trials ongoing).

3. Care in the Delivery Room

- **ILCOR** scientific review (2020) → AAP/AHA **North American** neonatal resuscitation guidelines.

Immediate care

- **Risk factors** that call for a resuscitation team at delivery:
 1. Poor maternal health
 2. Prenatal complications incl. suspected malformations
 3. **Preterm**
 4. Labor complications
 5. Long labor / **prolonged rupture of membranes**
 6. Anesthesia affecting placental perfusion
 7. Difficult delivery
 8. Labor medications (dose, route, timing)
- **Pre-delivery huddle:** expected GA, **amniotic fluid color**, extra risk factors, **cord-clamping plan**.
- **Causes of a nonvigorous newborn:** immaturity, hypoxemia/acidosis, sepsis, recent maternal drugs, **meconium-stained fluid**, CNS anomalies; respiratory: lung anomalies, upper airway obstruction, **pneumothorax**, meconium aspiration.

Umbilical cord management

- **Term:** delay **30–60 s** → ↑ Hb at birth, ↑ infant iron stores. Only harm: **hyperbilirubinemia / more phototherapy**.
- **Preterm:** ↓ **transfusion, IVH, NEC** and ↑ **survival**.
- **Delay for all preterm and term newborns not needing resuscitation.**
- **Don't delay** if resuscitation needed, or with **abruption, cord avulsion, bleeding previa or vasa previa**.
- **Cord milking:** reasonable alternative to immediate clamping for **late-preterm and term**; **avoid <28 wk**.

Pearl: Chapter 27 quotes organizations advising against routine milking **<29 wk**; this chapter says **<28 wk**. Either way, no milking in the extremely preterm.

Newborn resuscitation

- **~10%** need some active resuscitation; **~1%** need extensive care.
- Home birth neonatal death rate is **4×** attended hospital birth.

Apnea sequence (Fig 32-1)

Stage	Breathing	HR / BP / tone	Responds to stimulation?
Early O ₂ lack / CO ₂ rise	Transient rapid breathing	—	—
Primary apnea	Stops	HR falls , tone lost	Yes (simple stimulation)
Gasping	Deep, irregular gasps	—	—
Secondary apnea	Stops	HR falls further, BP falls , tone lost	No ; death unless ventilated

- **Primary and secondary apnea look the same clinically** → if no immediate response to stimulation, **start effective ventilation quickly**.
- **Effective PPV is the most critical intervention.**

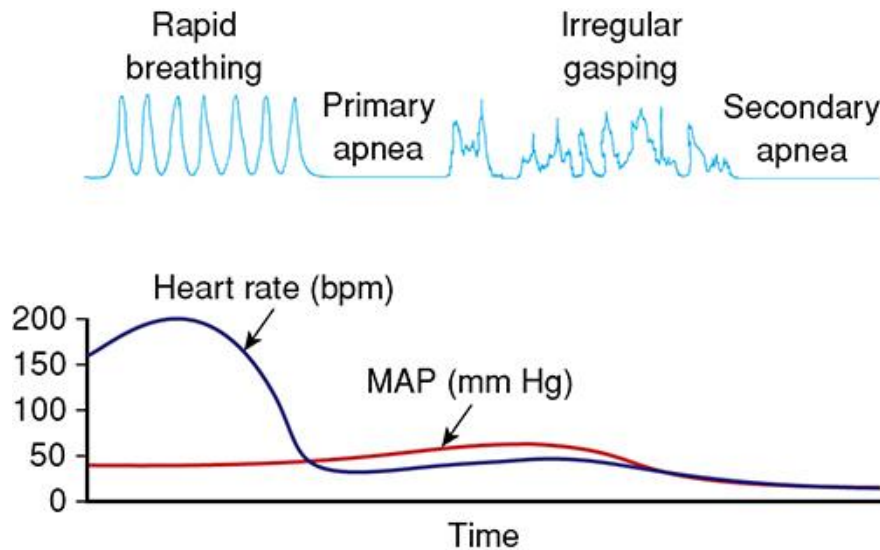


Figure 32-1. Rapid breathing → primary apnea → gasping → secondary apnea, with heart rate falling first and blood pressure falling in secondary apnea.

4. Resuscitation Protocol

Initial assessment (during delayed cord clamping)

- Assess **tone, breathing and heart rate**. Most term babies are **vigorous by 10–30 s**.
- **Vigorous**: stay with mother; skin-to-skin, dry, warm blanket. Keep **36.5–37.5°C**. **No routine oral suctioning**; bulb suction only for apnea or copious secretions. Dry, rub the back, observe.
- **Not vigorous or preterm** → **radiant warmer**: remove the wet blanket, dry.
- **Preterm thermal care**:
 - Delivery room **>25°C**
 - Plastic or wool **hat**
 - **Polyethylene wrap/"poncho"** (↓ evaporative loss)
 - **Chemically activated thermal mattress** (↓ conductive loss)
 - **Warm, humidified** respiratory gases
- Position with **mild neck extension**; suction mouth then nose only if apneic or obstructed.
- **⚠ Meconium**: **no routine intubation/suction** for the **nonvigorous** newborn; only for suspected **airway obstruction**. (Meconium still ↑ need for advanced resuscitation.)
- **Apnea, gasping or HR ≤100 bpm** after initial steps → **PPV with room air**, started **by 60 s** of life.

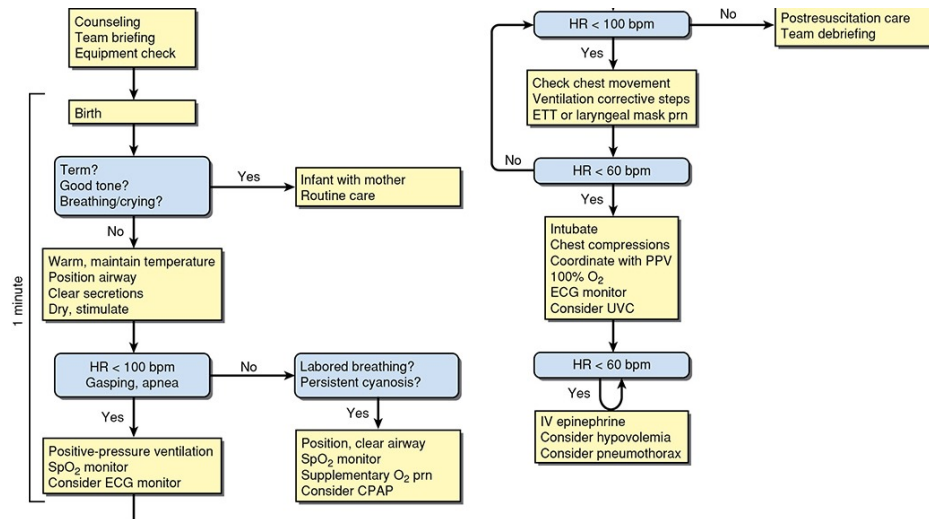


Figure 32-2. ILCOR/AAP/AHA newborn resuscitation algorithm (shown in two columns): initial steps within the first minute; PPV for HR <100 or apnea; compressions + 100% O₂ for HR <60; IV epinephrine if HR stays <60.

Note: the text uses ≤100 bpm and ≤60 bpm in places, whereas the algorithm uses <100 and <60 bpm.

Mask ventilation

- Rate **40–60 breaths/min**.
- **Pulse oximetry**; add O₂ in graded steps to meet the targets below.
- **Best sign of adequate ventilation: rising heart rate**. Colorimetric **ETCO₂** detector between device and mask confirms gas exchange.

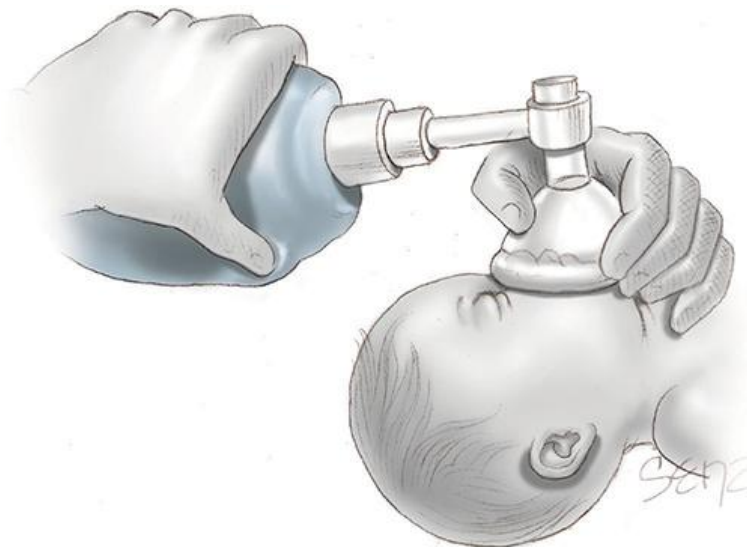


Figure 32-3. Bag-and-mask ventilation: sniffing position, nose to the ceiling, neck not hyperextended.

Table 32-1. Oxygen Saturation Goals per Minute of Life

Minute of life	Target SpO ₂
1 min	60–65%
2 min	65–70%
3 min	70–75%
4 min	75–80%
5 min	80–85%
10 min	85–95%

Memory hook: Start at **60%** and add **5% per minute** to 85% at 5 min; **85–95% by 10 min**.

- HR still **≤100 after 5–10 breaths** with inadequate ventilation → **MR. SOPA**. The two commonest problems are **mask leak** and **airway malposition**. Keep the **mouth open** (better conduit than the nares).

Table 32-2. Ventilation Corrective Steps (MR. SOPA)

Step	Action
M —Mask adjustment	Check the mask seal and reapply if needed
R —Reposition airway	Ensure the newborn is truly in the open airway position with mild neck extension
S —Suction mouth and nose	Remove obstructing secretions
O —Open the mouth	Avoid closing the mouth during efforts to achieve a good mask seal
P —Pressure increase	Increase the inflation pressure
A —Advanced airway	If all prior steps fail to achieve chest rise, then intubate or place a laryngeal mask or other supraglottic airway

Alternative airway

- For ineffective or prolonged mask ventilation.
- **Straight laryngoscope blade: size 0 preterm, size 1 term**. Gentle cricoid pressure may help.
- **Confirm tracheal placement: rising HR + ETCO₂ detection** (primary); also symmetric chest movement, equal breath sounds (axillae), none/gurgling over the stomach.
- ETT suction only for suspected obstruction. Attach self-inflating bag, flow-inflating bag or **T-piece**.

	Term	Preterm
Rate	40–60/min	40–60/min
Opening pressure	30–40 cm H ₂ O	20–25 cm H ₂ O
After lung inflation	20–25 cm H ₂ O	—

Chest compressions

- **HR <60 bpm despite ventilation corrective steps including an ETT**.
- Once the ETT is secured, compress **from the head of the bed** (leaves room for **umbilical venous access**).
- **↑ O₂ to 100%**.
- **Two-thumb, hands-encircling technique** on the **lower third of the sternum**, depth **~1/3 of the AP chest diameter** (enough for a palpable pulse). Less fatigue, higher perfusion pressure, less injury.
- **3:1 ratio: 90 compressions + 30 breaths = ~120 events/min**.
- Continue until spontaneous **HR ≥60 bpm**.

Epinephrine

- HR still **≤60** after adequate ventilation + compressions.

Route	Dose
IV (preferred)	0.01–0.03 mg/kg
Endotracheal (if no IV; less reliable)	0.05–0.1 mg/kg

Stopping resuscitation

- ILCOR: reasonable to stop if **no heartbeat after ≥20 min** of continuous, adequate resuscitation; **individualize**.
- Extended from 10 min because some hypothermia-trial babies with a **10-min Apgar of 0 survived without disability**.

5. Evaluation of Newborn Condition

Apgar score

- Dr. Virginia Apgar, **1953**. Five signs scored **0, 1 or 2**. Record at **1 and 5 min** in all.
- Score **<7** → repeat **every 5 min up to 20 min** (or until resuscitation stops). Expanded form also records **resuscitation interventions**.

Table 32-3. 20-Minute Expanded Apgar Score

Sign	0 point	1 point	2 point
Color	Blue or pale	Acrocyanotic	Completely pink
Heart rate	Absent	<100/min	>100/min
Reflex irritability	No response	Grimace	Cry or active withdrawal
Muscle tone	Limp	Some flexion	Active motion
Respiration	Absent	Weak cry; hypoventilation	Good, crying

The form has columns for scores at 1, 5, 10, 15 and 20 min plus a Total, and a Resuscitation section recording at each of those minutes: Oxygen, PPV/CPAP, ETT, Chest compressions, Epinephrine, with space for comments. CPAP = continuous positive airway pressure; ETT = endotracheal tube; PPV = positive-pressure ventilation.

Memory hook: APGAR = **A**pppearance (color), **P**ulse, **G**rimace (reflex irritability), **A**ctivity (tone), **R**espiration.

Prognostic value (Parkland, >150,000)

- Term, 5-min Apgar **7–10**: neonatal death **~1 in 5000**.
- Term, 5-min Apgar **≤3**: mortality **25%**.
- Equally predictive in preterms (also 5- and 10-min scores at 22–36 wk).
- **Apgar alone does NOT predict HIE**, but in HIE cohorts a low **10-min** score predicts school-age outcome.
- ⚠️ **Apgar alone cannot diagnose asphyxia or attribute cerebral palsy to hypoxia**. Confounders: **prematurity**, malformations, maternal drugs, infection.

Umbilical cord blood acid–base studies

- **Sampling**: isolate a **10–20 cm** cord segment with two clamps near the baby and two near the placenta; cut between each pair.
- Draw **arterial** blood into a **1–2 mL** heparinized syringe (lyophilized heparin, or flushed with **1000 U/mL**); cap and send **on ice**.
- **pH and Pco₂ stable at room temperature for up to 60 min**. Models can estimate birth values up to **60 h** later.
- Results can vary between analyzers.

Fetal acid–base physiology

Acid	Source	Clearance	Accumulation gives
Carbonic (H₂CO₃)	Oxidative metabolism (CO ₂)	Fast , via placenta	Respiratory acidemia (↓ CO ₂ exchange)
Organic (lactic, β-hydroxybutyric)	Anaerobic glycolysis (persistent exchange impairment)	Slow	Metabolic acidemia (↓ HCO ₃ [−] as it buffers)
Both	—	—	Mixed respiratory–metabolic acidemia

- In the fetus these form a **progressive continuum** (the placenta is lungs and, partly, kidneys). Usual cause: ↓ **uteroplacental perfusion** → CO₂ retention → if prolonged/severe, mixed or metabolic acidemia.

- **Henderson-Hasselbalch:** $\text{pH} = \text{pK} + \log [\text{base}]/[\text{acid}]$, i.e. $\text{pH} = \text{pK} + \log (\text{HCO}_3^- \text{ mEq/L}) / (\text{Pco}_2 \text{ mm Hg})$.
 HCO_3^- = metabolic component; Pco_2 = respiratory component.
- pH is **logarithmic**: the $[\text{H}^+]$ change from 7.0 $\rightarrow$ 6.9 is **almost twice** that from 7.3 $\rightarrow$ 7.2.
- **Delta base** is a more linear measure of metabolic acid: $\text{BE} = 0.02786 \times \text{Pco}_2 \times 10^{(\text{pH} - 6.1)} + 13.77 \times \text{pH} - 124.58$. Fig 32-4 gives it from any two values.
 - **Base deficit** = HCO_3^- below normal; base excess = above normal.
 - A mixed acidemia with a **large base deficit and low HCO_3^- (e.g., 12 mmol/L)** is more often associated with a **depressed neonate** than one with a near-normal HCO_3^- .

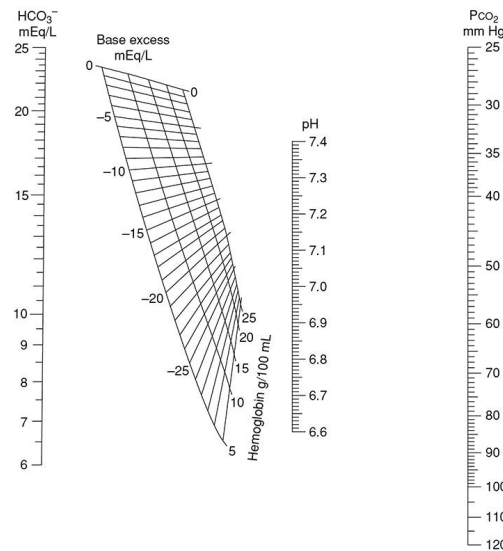


Figure 32-4. Siggaard-Andersen nomogram: a line through any two of HCO_3^- , pH and Pco_2 gives the base excess (by hemoglobin).

Clinical significance of acidemia

Table 32-4. Umbilical Cord Blood pH and Blood Gas Values in Normal Term Newborns

Values	Ramin, 1989 ^a Spontaneous Delivery n = 1292 ^c	Riley, 1993 ^b Spontaneous Delivery n = 3522 ^c	Kotaska, 2010 ^b Spontaneous Delivery n = 303 ^d	Kotaska, 2010 ^e Cesarean Delivery n = 189 ^d
Arterial Blood				
pH	7.28 (0.07)	7.27 (0.069)	7.26 (7.01–7.39)	7.3 (7.05–7.39)
Pco_2 (mm Hg)	49.9 (14.2)	50.3 (11.1)	51 (30.9–85.8)	54 (37.5–79.5)
HCO_3^- (mEq/L)	23.1 (2.8)	22.0 (3.6)	—	—
Base excess (mEq/L)	−3.6 (2.8)	−2.7 (2.8)	—	—
Venous Blood				
pH	—	7.34 (0.063)	7.31 (7.06–7.44)	7.34 (7.10–7.42)
Pco_2 (mm Hg)	—	40.7 (7.9)	41 (24.9–70.9)	44 (29.1–70.2)
HCO_3^- (mEq/L)	—	21.4 (2.5)	—	—
Base excess (mEq/L)	—	−2.4 (2)	—	—

^aNewborns of selected women with uncomplicated vaginal deliveries. ^bNewborns of unselected women with vaginal deliveries. ^cData shown as mean (SD). ^dData shown as range with 2.5 or 97.5 percentile. ^eCesarean delivery—labor not stated. From Centers for Disease Control and Prevention, 2012; Watson, 2006.

- Oxygenation and pH normally **fall during labor**. Preterm values are similar.
- **Lower limit of normal pH: 7.05–7.16 at 28–42 wk** (varies with GA) → defines **acidemia**.
- Most fetuses tolerate **pH as low as 7.00** without neurological injury.
- **pH <7.0 (Parkland)**: neonatal death **8%**, ICU admission **39%**, intubation **14%**, seizures **13%**. Neonatal encephalopathy **3%** (>51,000 term babies). Even with a normal 5-min Apgar, pH <7.0 → ↑ respiratory distress, NICU admission, sepsis.
- **Faster resolution** of acidemia → better outcome.

Respiratory acidemia

- Acute interruption of gas exchange, most often **transient cord compression**. **Generally not harmful**.
- **Correction rule**: subtract the upper-normal neonatal P_{CO_2} (~50 mm Hg); **each extra 10 mm Hg P_{CO_2} lowers pH by 0.08**.
- **Worked example (cord prolapse, cesarean 20 min later)**: pH **6.95**, P_{CO_2} **90 mm Hg** → $90 - 50 = 40$ mm Hg **excess CO_2** → $(40 \div 10) \times 0.08 = 0.32$ → $6.95 + 0.32 = 7.27$ → pre-prolapse pH was normal, so the acidemia was **respiratory**.

Metabolic acidemia

Definition	Criteria
Low (1997): fetal acidosis	Base deficit ≥ 12 mmol/L
Low (1997): severe fetal acidosis	Base deficit ≥ 16 mmol/L
Parkland (Casey, 2 SD below the mean)	UA pH <7.00 , $P_{CO_2} \leq 76.3$ mm Hg (higher = respiratory component), $HCO_3^- \leq 17.7$ mmol/L, base deficit ≥ 10.3 mEq/L
ACOG (possible neurological injury)	UA pH <7.0 and base deficit ≥ 12 mmol/L

- Linked to **multiorgan dysfunction**; rarely severe enough to cause **HIE**. Severe metabolic acidosis **poorly predicts** later neurological impairment at term, but **a fetus without acidemia cannot have had recent hypoxic injury**.
- **Preterms**: acid–base status is more closely tied to **IVH, PVL** and poorer long-term outcome.
- **Term neonates with metabolic acidemia + 5-min Apgar ≤ 3** : neonatal death **>3200×** that with Apgar ≥ 7 .

Recommendations for cord blood gases

- Some centers test **all** neonates (cost analysis suggests benefit and savings).
- **Reasonable to obtain for**:
 - Cesarean for **fetal compromise**
 - **Abnormal FHR tracing**
 - Intrapartum **fever**
 - **Low 5-min Apgar**
 - **Multifetal gestation**
 - **Severe FGR**
- Poor predictors of neurological outcome, but **the most objective evidence of fetal metabolic status at birth**.

6. Preventive Care

Eye infection prophylaxis (ophthalmia neonatorum)

- Mucopurulent conjunctivitis; commonest causes **gonococcus and chlamydia**. Gonococcal prophylaxis is **mandatory** in most states.

Situation	Management
Routine prophylaxis (all)	0.5% erythromycin ophthalmic ointment soon after delivery
Mother with untreated gonorrhea (presumptive neonatal GC conjunctivitis)	Ceftriaxone 25–50 mg/kg IV or IM, single dose, max 125 mg IM ; test for GC and chlamydia first
Chlamydial conjunctivitis	Oral erythromycin base/ethylsuccinate 50 mg/kg/d in 4 doses × 14 d , or oral azithromycin 20 mg/kg once daily × 3 d . Erythromycin only ~80% effective → a 2nd course may be needed; follow up

- Vaginal birth with active maternal chlamydia → **30–50%** conjunctivitis. **Topical prophylaxis doesn't reliably prevent chlamydial conjunctivitis** → rely on **prenatal screening/treatment**.
- Conjunctivitis **up to 3 months** of age → consider chlamydia.

Hepatitis B

- **Thimerosal-free vaccine within 24 h** for all medically stable newborns **>2000 g**.
- **Mother HBsAg-positive** → **also HBIG** (passive immunization). Some advocate maternal **antiviral nucleos(t)ide analogues** to cut transmission.

Vitamin K

- **0.5–1 mg IM, single dose, within 1 h of birth** → prevents **vitamin K deficiency bleeding**.

COVID-19

- Attendants: **airborne + contact precautions** (aerosolizing resuscitation).
- **~2%** of newborns of SARS-CoV-2-positive mothers test positive at **24–96 h**. Neonatal deaths rare; symptomatic maternal infection ↑ **preterm birth** and perinatal morbidity.
- Infected mother **masks** while holding the baby. Acutely ill → temporary separation. Not acutely ill → **rooming-in** with distance where possible; **breastfeeding with mask + hand hygiene** is fine.
- **Test the newborn at least once before discharge**.

Newborn screening

- **35 core conditions** (Advisory Committee on Heritable Disorders in Newborns and Children); most states require the full core panel, some add secondary conditions (see babysfirsttest.org).

Table 32-5. Newborn Screening Core Panel

Organic Acid Metabolism	Fatty Acid Metabolism	Amino Acid Metabolism	Others
Isovaleric acidemia	Medium-chain acyl-CoA dehydrogenase deficiency	Phenylketonuria	Biotinidase deficiency
Propionic acidemia	Very long-chain acyl-CoA dehydrogenase deficiency	Maple syrup urine disease	Galactosemia
Glutaric acidemia type I	Long-chain 3-OH acyl-CoA dehydrogenase deficiency	Homocystinuria	Hearing loss
β-Ketothiolase deficiency	Trifunctional protein deficiency	Citrullinemia, type I	Cystic fibrosis
3-Hydroxy-3-methylglutaric aciduria	Carnitine uptake defect	Argininosuccinic	Critical congenital heart disease
Holocarboxylase synthase deficiency	Endocrine Disorders:	Tyrosinemia, type I	Severe combined immunodeficiency
3-Methylcrotonyl-CoA carboxylase deficiency	Congenital hypothyroidism	Hemoglobin Disorders:	Glycogen storage disease, type II
Methylmalonic acidemia (methylmalonyl-CoA mutase)	Congenital adrenal hyperplasia	SS disease	Mucopolysaccharidosis, type 1
Methylmalonic acidemia (cobalamin disorder)		S-β-thalassemia	X-linked adrenoleukodystrophy
		SC disease	Spinal muscular atrophy

From Advisory Committee on Heritable Disorders in Newborns and Children, 2018; Watson, 2006. (9 organic acid + 5 fatty acid + 2 endocrine + 6 amino acid + 3 hemoglobin + 10 others = 35 conditions.)

7. Routine Newborn Care

Gestational age estimation

- Done soon after birth. **SGA and LGA** neonates → ↑ **hypoglycemia and polycythemia** → check **glucose and hematocrit**.

Skin and umbilical cord

- Wipe off excess vernix, blood, meconium; remaining vernix absorbs within **24 h**. **First bath after temperature is stable**.
- **Dry cord care is sufficient** (AAP); **no dressing** (air speeds drying and separation).
 - Stump dries and blackens within **24 h**; usually **separates within 2 weeks**.
 - Resource-poor settings / high infection rates: local antimicrobials; **WHO: chlorhexidine**.
- **Omphalitis**: *S. aureus*, **group A and B streptococci**, gram-negatives (*E. coli*). Cellulitis and discharge; mild erythema/bleeding at separation is common; some have no outward signs.

Feeding and weight loss

- **Exclusive breastfeeding for 6 months**. US 2020: **84%** initiated, **58%** at 6 months, **35%** at 1 year.
- Term: **8–12 feeds/day**, ~**15 min** each; preterm/FGR need shorter intervals.
- Little intake for the first **3–4 days** → weight loss. **Term babies regain birthweight by ~10 days**; preterms lose more and regain more slowly; healthy FGR babies regain faster than preterms.

Stools and urine

- **Meconium** (2–3 days): desquamated cells, mucus, epidermal cells, **lanugo**; color from **bile pigments**. Gut sterile until hours after birth.
- **Meconium passed by 24 h in 90%**, most of the rest by **36 h**. First void soon after birth, possibly not until **day 2**.
- **No stool/urine by these times** → consider **Hirschsprung disease, imperforate anus, posterior urethral valves**.
- After day **3–4**: light-yellow, soft milk stools.

Neonatal hyperbilirubinemia

- **Physiological jaundice** in $\sim\frac{1}{3}$ between **days 2 and 5**; important with early discharge. Use AAP guidelines by GA, hour of life and risk factors (Ch 33).

Male circumcision

- US rate **57% (2003)** → **52% (2016)**.
- **Benefits**: prevents **phimosis, paraphimosis, balanoposthitis, UTI**; ↓ **penile cancer**, ↓ **cervical cancer** in partners, ↓ some **STIs incl. HIV**.
- **AAP Task Force (2012)**: benefits **outweigh risks** → access justified, but **not recommended for all**.
- **Contraindications**: unhealthy neonate, **genital anomalies (e.g., hypospadias)**, **family history of bleeding disorder** (unless excluded).
- **Analgesia required: dorsal penile nerve block or ring block > topical EMLA**; **sucrose pacifier** as adjunct.
 - **Ring block**: wheal of **1% lidocaine** at the penile base, needle advanced in a 180-degree arc to each side. **Max lidocaine 1.0 mL**.
 - ⚠ **Never add epinephrine** (ischemia).
- **Devices**: **Gomco**, **Mogen** (opens to **3 mm** max; quicker and less apparent discomfort than Gomco), **Plastibell**.
- **Technique goals**: remove enough shaft skin and inner prepuce to expose the glans (prevent phimosis):
 1. Estimate the external skin to remove accurately
 2. **Dilate the preputial orifice** to see a normal glans
 3. **Free the inner preputial epithelium** from the glans
 4. Leave the device on **long enough for hemostasis** before excising
- **Complications**: bleeding, infection, hematoma (low). Rare: **glans amputation**, HIV/STI transmission, **meatal stenosis**, penile denudation, destruction from **electrosurgery**, inclusion cyst, **urethrocutaneous fistula**, ischemia from lidocaine **with epinephrine**.



Figure 32-5. Circumcision tools: Mogen clamp (A; opens to 3 mm), Gomco clamp (B), Plastibell (C).

Rooming-in and discharge

- Rooming-in fosters bonding and **breastfeeding**; mother usually **fully ambulatory by 24 h**.
- WHO minimum stay 24 h**. Most newborns can go within **48 h**, but not all:
 - Belgium (stays cut by 1 day): ↑ readmission for **jaundice and dehydration**.
 - Washington: discharge **<30 h** → **28-day mortality 4×**, 1-year mortality **2×**.
 - Late-preterm** newborns need special care at discharge.
- Newborns' and Mothers' Health Protection Act (1996)**: insurers can't limit stays to **<2 days (vaginal)** or **<4 days (cesarean)**. California: readmissions fell **9, 12, 20%** at 1, 2 and 3 years.

Quick-Recall: Neonatal Resuscitation Triggers and Actions

Trigger	Action
Term, good tone, breathing/crying	Stay with mother , routine care (dry, warm, skin-to-skin)
Not vigorous or preterm	Radiant warmer: warm, position (mild extension), clear if needed, dry, stimulate
Apnea, gasping or HR <100 (by 60 s)	PPV (room air), 40–60/min , SpO ₂ monitor ± ECG
Labored breathing / persistent cyanosis (HR ≥100)	Position/clear airway, SpO ₂ , O₂ as needed , consider CPAP
HR still <100 after 5–10 breaths	MR. SOPA ; ETT or laryngeal mask
HR <60 despite effective ventilation (ETT)	Chest compressions 3:1, 100% O₂ , ECG, consider UVC
HR still <60	IV epinephrine 0.01–0.03 mg/kg (ETT 0.05–0.1 mg/kg); consider hypovolemia, pneumothorax
No heartbeat after ≥20 min	Reasonable to stop (individualize)
Meconium, nonvigorous	No routine intubation/suction

Key Numbers to Memorize

- Resuscitation needed: **10%** any, **1%** extensive
- Aeration within **3–5 breaths**; adult-like mechanics by **~5th breath**
- Delayed clamping **30–60 s**; no milking **<28 wk**
- Temperature **36.5–37.5°C**; preterm delivery room **>25°C**
- PPV by **60 s**; rate **40–60/min**; pressures **30–40** then **20–25** cm H₂O (term), **20–25** (preterm)
- SpO₂ targets **60–65% (1 min) → 80–85% (5 min) → 85–95% (10 min)**
- Laryngoscope blade **0** preterm, **1** term
- Compressions at **HR <60**; depth **1/3 AP**; **3:1, 90 + 30 = 120/min**; stop at **HR ≥60**
- Epinephrine **0.01–0.03 mg/kg IV, 0.05–0.1 mg/kg ETT**; stop after **20 min** asystole
- Apgar at **1, 5**, then q5 min to **20 min** if <7; term 5-min ≤3 → **25%** mortality; 7–10 → **1/5000**
- Cord segment **10–20 cm**; gases stable **60 min** at room temp
- Normal UA pH **~7.28**, Pco₂ **~50**, HCO₃⁻ **~23**, BE **~-3.6**; lower limit pH **7.05–7.16**
- pH <7.0: death **8%**, NICU **39%**, intubation **14%**, seizures **13%**, encephalopathy **3%**
- Pco₂ rule: each **10 mm Hg over 50** → **pH -0.08**
- Metabolic acidosis (ACOG): **pH <7.0 + BD ≥12**; severe **BD ≥16**
- Erythromycin eye ointment **0.5%**; ceftriaxone **25–50 mg/kg, max 125 mg**; chlamydia erythromycin **50 mg/kg/d × 14 d** or azithromycin **20 mg/kg × 3 d**
- Hep B vaccine **<24 h** if **>2000 g**; vitamin K **0.5–1 mg IM within 1 h**
- Newborn screen **35** core conditions

- Meconium by **24 h (90%)** / 36 h; birthweight regained by **10 days**; jaundice $\frac{1}{3}$ on days **2-5**; cord separates by **2 wk**
- Circumcision lidocaine max **1.0 mL**, no epinephrine; Mogen **3 mm**
- Discharge <30 h → **4x** 28-day mortality; insured stay **2 d** vaginal / **4 d** cesarean